

Documentation of Udemy Clone

Project Name:	Udemy Clone
Project Type:	Full-Stack Web Application (MERN stack)
Version:	1.0.0
Status:	In Development
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Description

Udemy Clone is a full-stack web app that works like the real Udemy website. It lets people search for courses, buy them, and start learning online. Teachers can create and manage their own courses, and students can watch lessons and see how much they've completed.

Purpose and Scope

The purpose of this project is to develop a scalable and secure e-learning platform, similar to Udemy, where:

- Instructors can create, manage, and sell their courses.
- Students can register, enroll in courses, and track their progress.

- Admins can monitor platform activity and manage users/content.

Objectives

- Deliver a multi-role application (Admin, Instructor, Student).
- Allow seamless content upload (videos).
- Implement secure authentication and payment.
- Provide course progress tracking, reviews, and certification .

System Architecture

- Frontend : React.js
- Backend : Node.js, Express.js
- Database : MongoDB

Technology Stack

Layer	Tech Used
Frontend	React.js, Axios, React Router
Backend	Node.js, Express.js
Database	MongoDB
Authentication	JWT, Bcrypt
Payment	stripe
File Storage	AWS S3
UI Styling	Bootstrap / TailwindCSS / Custom CSS

Modules

1. User Module

- Register/Login/profile
- Role management (Admin/Instructor/Student)
- Profile management

2. Course Module

- Course creation (title, description, pricing, video upload)
- Course editing and deletion
- Course listing and detail view

3. Enrollment Module

- Enroll in free/paid courses
- Watch course content
- Track progress

4. Payment Module

- Integration with Stripe/Razorpay
- Payment webhook processing
- Enrollment after successful payment

5. Admin Module

- User & course management
- Dashboard for analytics (future)

Database Schema

User

- id, name, email, password, role, joinedAt, other

Course

- id, title, description, price, instructorId, content[], createdAt, other

Enrollment

- id, studentId, courseId, enrolledAt, status, other

Review

- id, userId, courseId, rating, comment, other

API

Method	Endpoint	Description
POST	/api/auth/register	User registration
POST	/api/auth/login	User login
PUT	/api/auth/user	Update profile
POST	/api/courses	Create a new course
PUT	/api/courses	Update Course
GET	/api/courses	Get all courses
GET	/api/courses/:id	Get course by ID

POST	/api/enroll/:courseId	Enroll in a course
POST	/api/payment/create	Initiate payment
POST	/api/payment/webhook	Handle payment confirmation

Functional Requirements

- Users can register/login securely.
- Instructors can create/edit/delete courses.
- Students can enroll and access purchased courses.
- Secure payments must be processed via payment gateway.
- Admin can deactivate users and courses.

Non-Functional Requirements

- The application must support at least 10,000 concurrent users.
- Average response time for APIs should be < 300ms.
- Frontend must be mobile-responsive.
- All sensitive data must be encrypted .

Testing Plan

Test Type	Tools	Scope
Unit Testing	Jest, Mocha	Backend route
Integration Testing	Postman	API endpoints

UI Testing	React Testing Lib	UI components rendering
Manual Testing	Browser + Postman	End-to-end user flows

Deployment Plan

- **Frontend**

hosted on Vercel / Netlify

- **Backend**

Render/ VPS

- **Database**

MongoDB Atlas

- **CI/CD:**

GitHub + deployment hooks

Security Measures

- Passwords hashed using Bcrypt
- JWT-based authentication
- Role-based authorization
- Rate limiting and input validation
- HTTPS enforced in production

Maintenance Plan

- Monthly updates for dependencies
- Regular backups of MongoDB
- Add test cases as features grow

Future Enhancements

- Live classes
- Certificate generation
- Mobile app version
- Advanced analytics dashboard for instructors and admins